

SALES PERFORMANCE ANALYSIS DASHBOARD

PROJECT DOCUMENTATION

Submitted By: L.Thasneem

Course: AI-driven Data Analytics

Tool Used: Microsoft Power BI

PROJECT OVERVIEW

This project leverages **Power BI** to analyze and visualize sales data across multiple categories—Furniture, Clothing, and Electronics—using datasets of orders, detailed transactions, and monthly sales targets. The primary objective is to apply **DAX (Data Analysis Expressions)** for creating calculated measures such as total sales, profit margins, and category-wise performance, while comparing actual results against predefined targets.

OBJECTIVES OF THE PROJECT

The objective of this project is to analyze and visualize sales performance using **Microsoft Power BI**. It aims to demonstrate how data from multiple sources—**List of Orders**, **Order Details**, and **Sales Targets**—can be integrated, cleaned, and modeled to generate meaningful insights.

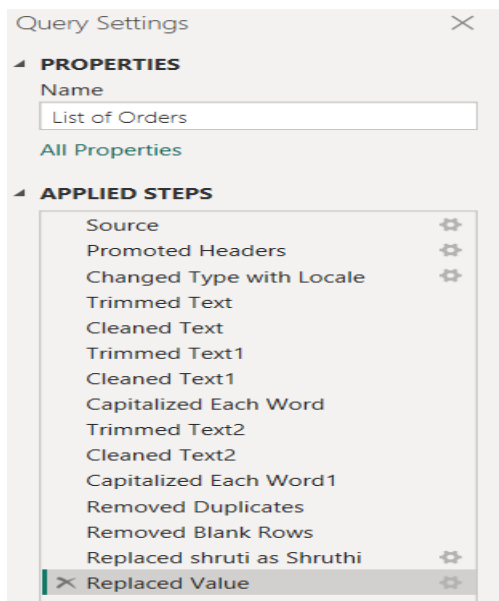
DATASETS USED

- a. List of Orders
- b. Order Details
- c. Sales Target

DATA PREPARATION AND DATA CLEANING

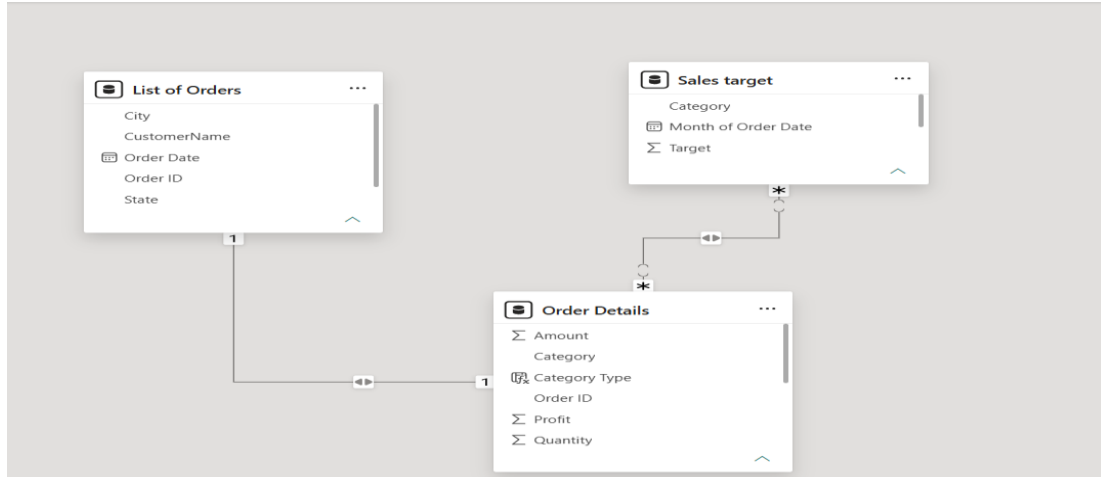
The data preparation process involved importing three datasets — *List of Orders*, *Order Details*, and *Sales Target* — into **Power BI**. Each dataset was cleaned and transformed using **Power Query** to ensure consistency and accuracy before modeling.

- **Duplicate Removal:** Checked and eliminated duplicate *Order IDs* to maintain data integrity.
- **Text Cleanup:** Applied TRIM and CLEAN functions to remove extra spaces and formatting inconsistencies in *CustomerName*, *State*, and *City* columns.



DATA MODELING AND RELATIONSHIPS

After cleaning and preparing the datasets, the next step was to build a **data model** in Power BI to establish logical connections between tables. This ensures smooth data flow and accurate cross-table analysis.



CALCULATED COLUMNS

Calculated columns were created using **DAX (Data Analysis Expressions)** to derive new insights from the existing data. These columns help in performing advanced analysis and building dynamic visualizations.

Category Type

```
1 Category Type = 'Order Details'[Category] & "-" & 'Order Details'[Sub-Category]
```

Amount	Profit	Quantity	Category	Sub-Category	Category Type
53	1	4	Clothing	Stole	Clothing-Stole
55	-39	4	Clothing	Stole	Clothing-Stole
133	12	5	Clothing	Stole	Clothing-Stole
27	-25	2	Clothing	Stole	Clothing-Stole
74	29	3	Clothing	Stole	Clothing-Stole
27	-15	1	Clothing	Stole	Clothing-Stole
17	6	1	Clothing	Stole	Clothing-Stole

Revenue per Order

The **Revenue per Order** column was created using **DAX** to calculate the total revenue generated from each order based on the *Order Details* dataset. This helps in understanding the monetary contribution of individual orders and identifying high-value transactions.

```
1 Revenue Per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]
```

Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order
53	1	4	Clothing	Stole	Clothing-Stole	₹ 212
55	-39	4	Clothing	Stole	Clothing-Stole	₹ 220
133	12	5	Clothing	Stole	Clothing-Stole	₹ 665
27	-25	2	Clothing	Stole	Clothing-Stole	₹ 54
74	29	3	Clothing	Stole	Clothing-Stole	₹ 222
27	-15	1	Clothing	Stole	Clothing-Stole	₹ 27
17	6	1	Clothing	Stole	Clothing-Stole	₹ 17
31	-11	4	Clothing	Stole	Clothing-Stole	₹ 124
8	-6	1	Clothing	Stole	Clothing-Stole	₹ 8

Sales Category

This analysis visualizes total sales across the three major categories — **Furniture**, **Clothing**, and **Electronics** — to understand their contribution to overall revenue. Using DAX measures and Power BI visuals, category-wise performance was evaluated monthly and compared against targets.

```

1 Sales Category = IF(
2     'Order Details'[Amount]> CALCULATE
3     (AVERAGE('Order Details'[Amount]),ALL('Order Details')),"Above Average","Below Average")
4
5

```

Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order	Sales Category
53	1	4	Clothing	Stole	Clothing-Stole	₹ 212	Below Average
55	-39	4	Clothing	Stole	Clothing-Stole	₹ 220	Below Average
133	12	5	Clothing	Stole	Clothing-Stole	₹ 665	Below Average
27	-25	2	Clothing	Stole	Clothing-Stole	₹ 54	Below Average
74	29	3	Clothing	Stole	Clothing-Stole	₹ 222	Below Average
27	-15	1	Clothing	Stole	Clothing-Stole	₹ 27	Below Average
17	6	1	Clothing	Stole	Clothing-Stole	₹ 17	Below Average

DAX MEASURES

DAX (Data Analysis Expressions) measures were created to perform dynamic calculations and aggregations across the data model.

These measures enable interactive analysis and visualization of key business metrics in Power BI.

Order Count

Order Count = `COUNT('Order Details'[Order ID])`

500

Order Count

Average Profit in Delhi

Average Profit in Delhi = `CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[City]="Delhi")`

₹ 43.7647

Average Profit in Delhi

Year-to-Date (YTD) Sales

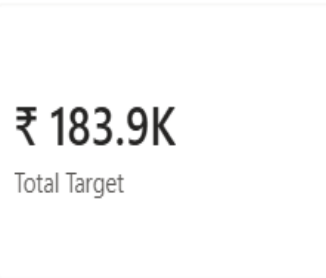
Year To Date Sales = `TOTALYTD(SUM('Order Details'[Amount]),'List of Orders'[Order Date])`



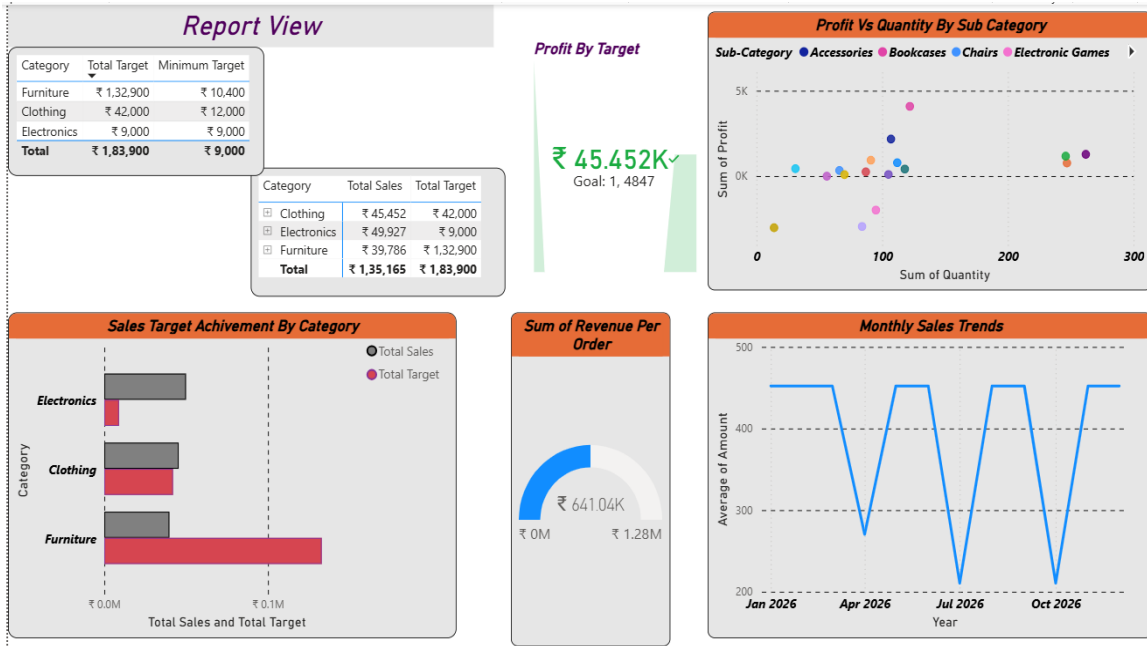
Total Sales
Total Sales = `SUM('Order Details'[Amount])`



Total Target
Total Target = `SUM('Sales target'[Target])`



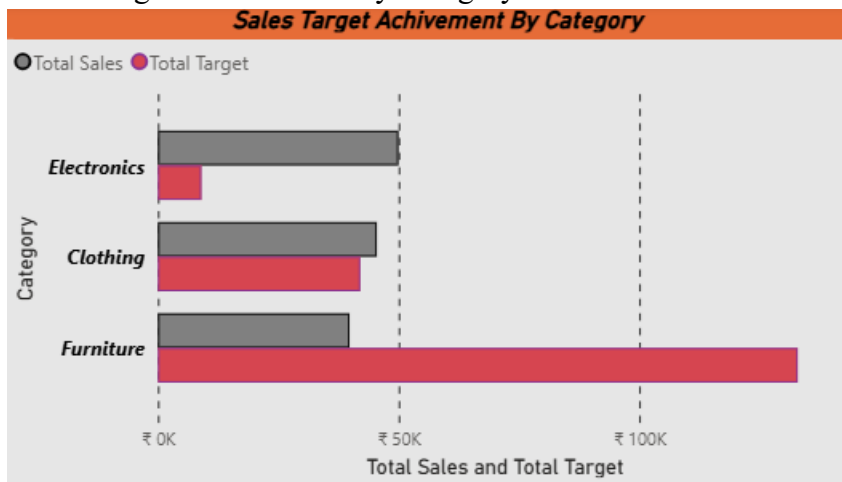
VISUALIZATIONS CREATED



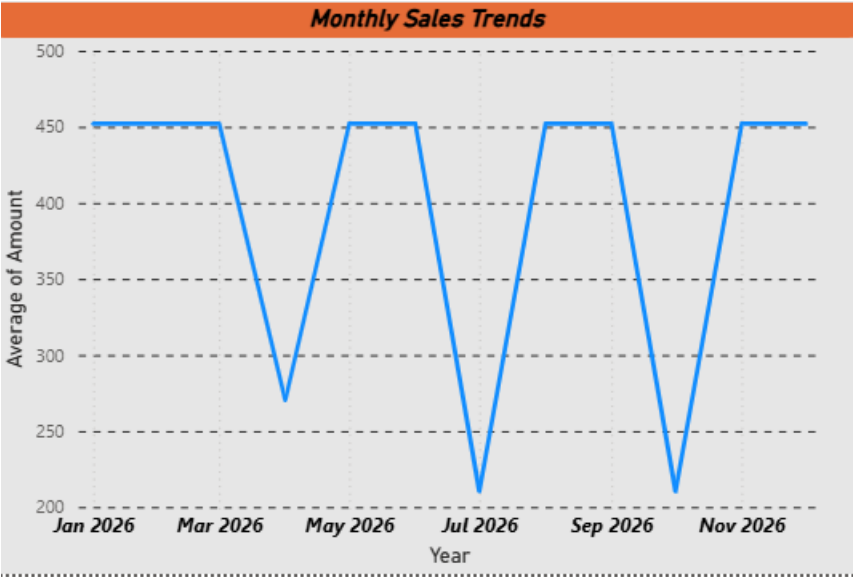
Total Target Card / Total Sales Card

Category	Total Sales	Total Target
Clothing	₹ 45,452	₹ 42,000
Electronics	₹ 49,927	₹ 9,000
Furniture	₹ 39,786	₹ 1,32,900
Total	₹ 1,35,165	₹ 1,83,900

Sales Target Achievement by Category



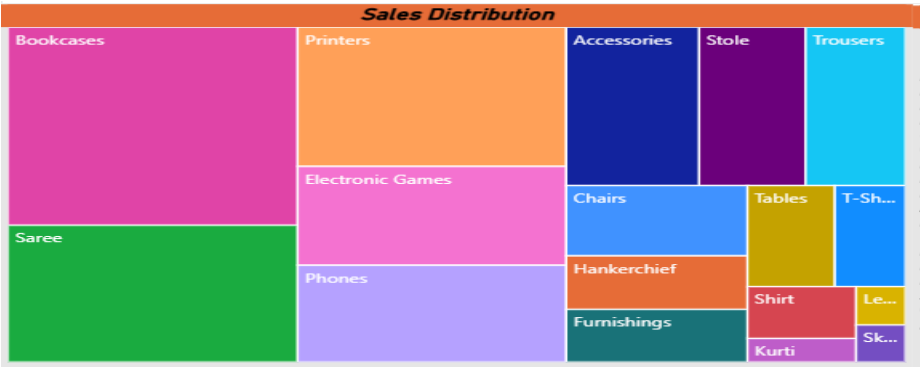
Monthly Sales Trend



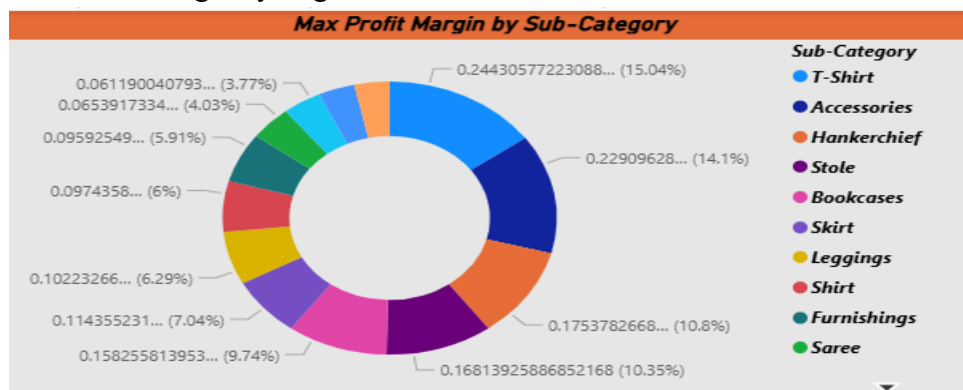
Geographic Sales Analysis



Sales Distribution by Sub-Category



Maximum Target by Segment



VISUALIZATION SCREENSHOTS AND EXPLANATIONS

Sales Target Achievement by Category

- This **horizontal bar chart** compares *Total Sales* and *Total Target* for **Electronics**, **Clothing**, and **Furniture**.
- **Furniture** shows the highest sales and target values, indicating strong revenue generation.
- **Clothing** performs moderately, while **Electronics** records the lowest figures, suggesting potential improvement areas.

Monthly Sales Trends

- The **line graph** displays the *Average of Amount* over time from **January 2026** to **November 2026**.
- The trend shows periodic fluctuations, with noticeable peaks around ₹450K and dips near ₹200K.
- These variations may reflect seasonal demand or promotional cycles.

Max Profit Margin by Sub-Category

- The **donut chart** illustrates the maximum profit margin percentages for various sub-categories.
- The top contributors are **T-Shirt (15.04%)**, **Accessories (14.1%)**, and **Shirt (10.8%)**, indicating these items yield the highest profitability.
- Sub-categories like **Bookcases** and **Furnishings** have smaller shares, showing lower margins.

Profit vs Quantity by Sub-Category

- The **scatter plot** compares *Sum of Profit* and *Sum of Quantity* across sub-categories such as **Accessories**, **Chairs**, **Handkerchief**, and **Leggings**.

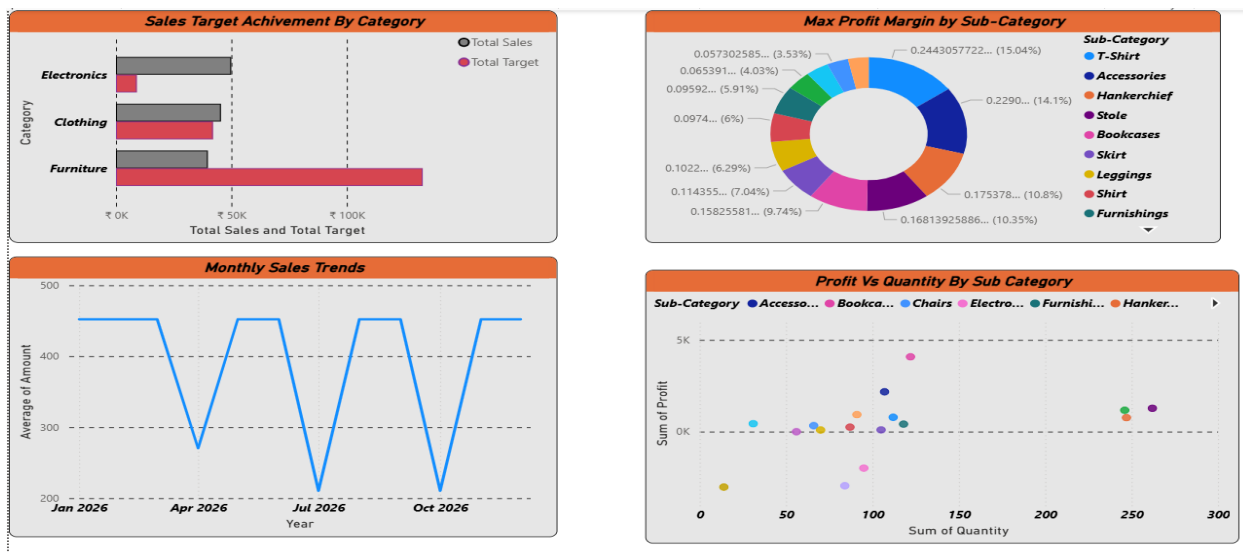
- Each colored data point represents a sub-category, helping identify which products generate higher profits relative to quantity sold.
- For instance, **Accessories** and **Bookcases** show high profit levels, while **Handkerchiefs** and **Leggings** have lower profitability.

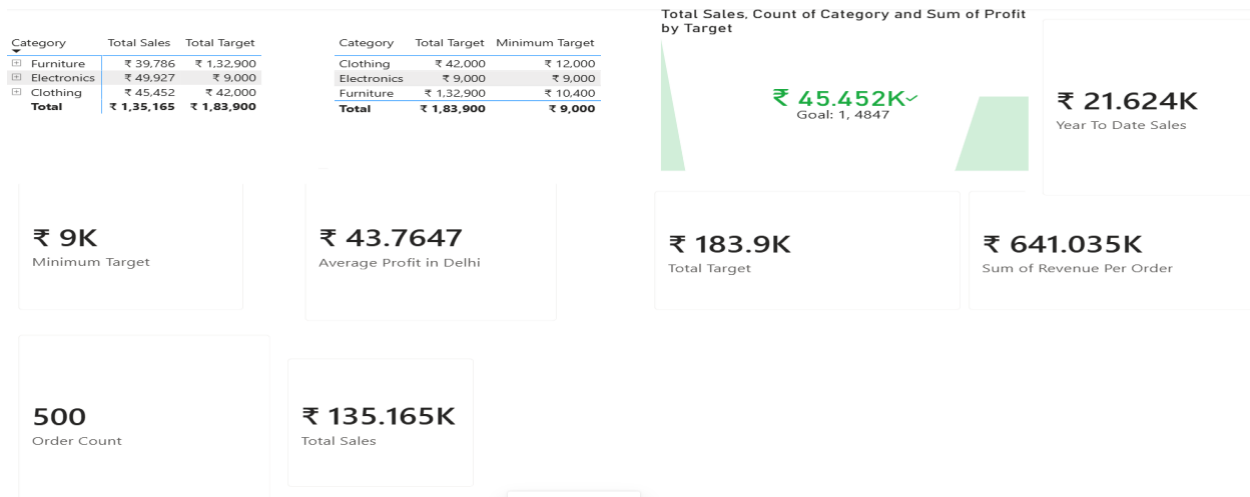
Gauge Chart – Overall Sales Performance

- The **gauge visual** displays total sales value, approximately ₹641.04K out of ₹1.28M.
- This indicates that overall sales have achieved nearly half of the expected target, suggesting room for growth in upcoming months.

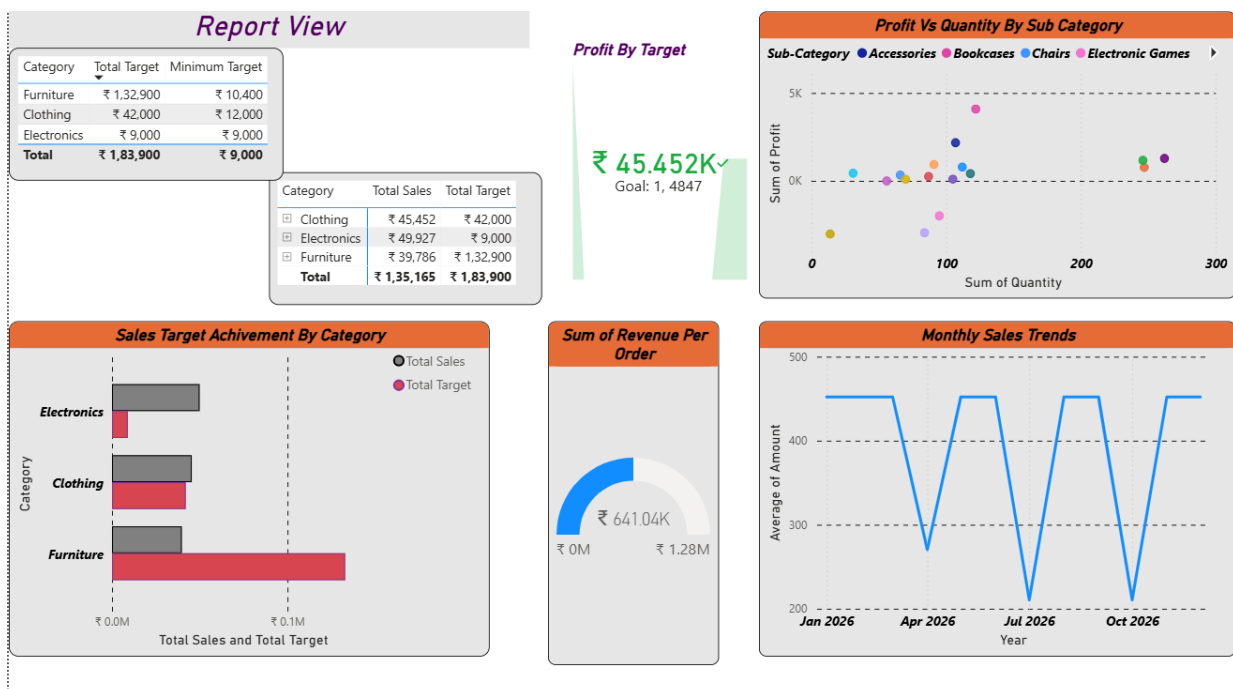
Order Count by State

- **Madhya Pradesh** leads with **55 orders**, making it the top-performing state in terms of order volume.
- **Maharashtra (19 orders)** and **Punjab (16 orders)** follow, indicating strong engagement in these regions.
- States like **Karnataka, Uttar Pradesh, Gujarat, and Rajasthan** show moderate order counts ranging between 10–15.
- Several other states — including **Delhi, Kerala, Andhra Pradesh, Bihar, and Tamil Nadu** — have fewer orders, suggesting opportunities for targeted marketing or expansion.





DASHBOARD LAYOUT SCREENSHOT



KEY INSIGHTS

The Power BI dashboard reveals several important insights derived from the integrated datasets (*List of Orders, Order Details, and Sales Target*)

Category Performance

- **Furniture** achieved the highest total sales but fell short of its monthly targets in several periods.
- **Clothing** maintained consistent growth and frequently met or exceeded targets, indicating strong customer demand.
- **Electronics** showed fluctuating performance, with notable peaks during promotional months.

Profitability Trends

- Profit margins varied significantly across categories — *Clothing* contributed the most to overall profit, while *Furniture* had high revenue but lower margins.
- Certain sub-categories like *Printers* and *Bookcases* generated high sales but moderate profits, suggesting potential pricing adjustments.

Regional Insights

- Major sales concentrations were observed in **Maharashtra, Madhya Pradesh, and Rajasthan**, highlighting these as key markets.
- Cities like **Mumbai, Indore, and Jaipur** consistently ranked among top performers.

Customer Behavior

- Repeat orders and high-value transactions were more frequent in metropolitan regions.
- Seasonal spikes in April and December suggest promotional or festive influences on buying patterns.

CONCLUSION

This Power BI project successfully demonstrates how **data analysis and visualization** can transform raw datasets into meaningful business insights. By integrating *List of Orders*, *Order Details*, and *Sales Target* data, the dashboard provides a clear view of sales performance across categories, regions, and time periods.